Lab 1 - AI^2 Product Description

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1 Introduction

In the era of automation, approximately 60% of companies still rely on Accounts Payable (AP) manual processing for handling invoices (Tamaro, 2025). Manual processing carries a variety of disadvantages that make the task expensive, slow, and prone to costly errors. These issues can have a negative effect on a company as a whole, its individual workers, and the vendors that a company conducts business with.

Due to the excessive labor hours, printing, and mailing expenses, each manually processed invoice can cost between $12 and $40 (Gunnoo, 2025). When automated, those same invoices can be processed at a cost between $3 and $6 (Gunnoo, 2025), which adds up quickly for a company processing tens of thousands of purchases per year. Manual labor hours account for 62% of total AP processing costs, most of which go into checking simple, repetitive invoices that have potential for automation (Tamaro, 2025).

Manual processing is inefficient, with an invoice taking 10 to 15 days to process, while an automated system can process the same invoice in under 5 days (Jex, 2024). Potential for human error can cause processing to take even longer. Even if no errors occur, this workload stifles the company's potential as a whole. The AP team being pushed to their limit by the current workload makes it difficult or unfeasible for the company to take on new clients, limiting the company's potential for growth.

Manual processing comes with high error potential. 39% of invoices contain errors, and a third of companies make duplicate payments due to such errors (Jex, 2024). 59% of accountants make several financial errors monthly, many of whom cite overexertion at work and insufficient processing tools available (Gartner, 2024). These many errors can lead to further delays, strained relationships with vendors, and financial loss for the company.

The issues of AP manual processing are magnified at ADS, which handles over 150,000 invoices per year, arriving from about 3,000 suppliers in a variety of formats. Their current AP team consists of only 6 people, who must manually review each invoice and compare it to the company's purchase orders. Each team member must handle an average of 50 to 100 invoices per business day, many of which are simple and repetitive. The end result is a high workload, increased potential for errors and delays, and an inability to take in more work on top of the existing load.

A solution to these problems should aim to lighten the AP team's workload, allowing them to do more skillful and meaningful work instead of repetitive checking, and improve employee morale. This shift to automation can lead to an 80% increase in productivity for the AP team (Kearns, 2025). The company as a whole would benefit from more efficient financial processes and greater potential for growth. Vendors would benefit from having their payments processed quickly and efficiently.

AI^2 is a web-based platform that seeks to alleviate these problems by managing high volumes of invoices quickly and accurately. It is designed to streamline repetitive invoice processing and allow AP team members to instead focus on high-risk invoices that demand greater scrutiny. It automates invoice intake and matching, while routing edge cases for staff review, reducing backlogs without removing human oversight. This allows the company to accelerate payment cycles, strengthen supplier relationships, and improve overall financial health.

2 Product Description

AI^2 is a web-based application designed to help organizations manage high volumes of invoices with greater speed, efficiency, and transparency. It contains features for use by AP team

members, AI model trainers, administrators, and compliance officers. These features work together to streamline all aspects of AP processing, particularly allowing teams to handle high volumes by automating easy, high-confidence matches and prioritizing low-confidence ones.

2.1 Key Product Features and Capabilities

The most critical features of AI^2 for AP team members are its dashboard and matching algorithm. They are able to view all new invoices in one central dashboard as they are pulled from e-mail feeds and attachments by the e-mail parsing API. The matching algorithm will attempt to match each invoice to a purchase order and return a confidence score. Matches will be flagged based on a given score threshold, notifying the AP team to manually review the data and make corrections as needed. Invoices can be assigned to specific team members based on factors such as the supplier or the existing assignment queues. As corrections are made, they are logged so that the AI model trainers can make AI^2 more accurate in the future.

AI model trainers are able to give AI^2 anonymized training data to build its matching capabilities and then deploy completed models. Additional data can be anonymized and added to the system to retrain and improve the algorithm's accuracy. The invoice quality analyzer notifies when the e-mail parsing API fails to properly extract the required data fields so that adjustments can be made. Model trainers can view AI^2's performance history to ensure its output aligns with company expectations.

Administrators are able to view and adjust AI^2's configurations, such as confidence score thresholds, routing rules, and notification preferences, to match the company's changing policies and workflows. They can assign invoices to specific AP team members based on supplier names, and then rebalance these assignments as needed to keep the workload spread efficiently. They can authorize AI^2 to connect to an Outlook inbox for retrieving incoming invoices, and to Oracle

NetSuite for processing and connection to the database. AI^2 will also notify administrators if it fails to integrate with existing computing systems or invoices are not properly routed. Immutable audit logging for all user and system actions will allow administrators to ensure that AI^2 adheres to government security frameworks and can pass compliance checks.

Compliance officers are able to perform regular reviews of AI^2's audit logs to ensure that they comply with company policies and external regulatory requirements. To this end, invoice and user data is encrypted both at rest and in transit. All configuration and policy changes are also logged to prevent any unauthorized adjustments.

2.2 Major Components (Hardware/Software)

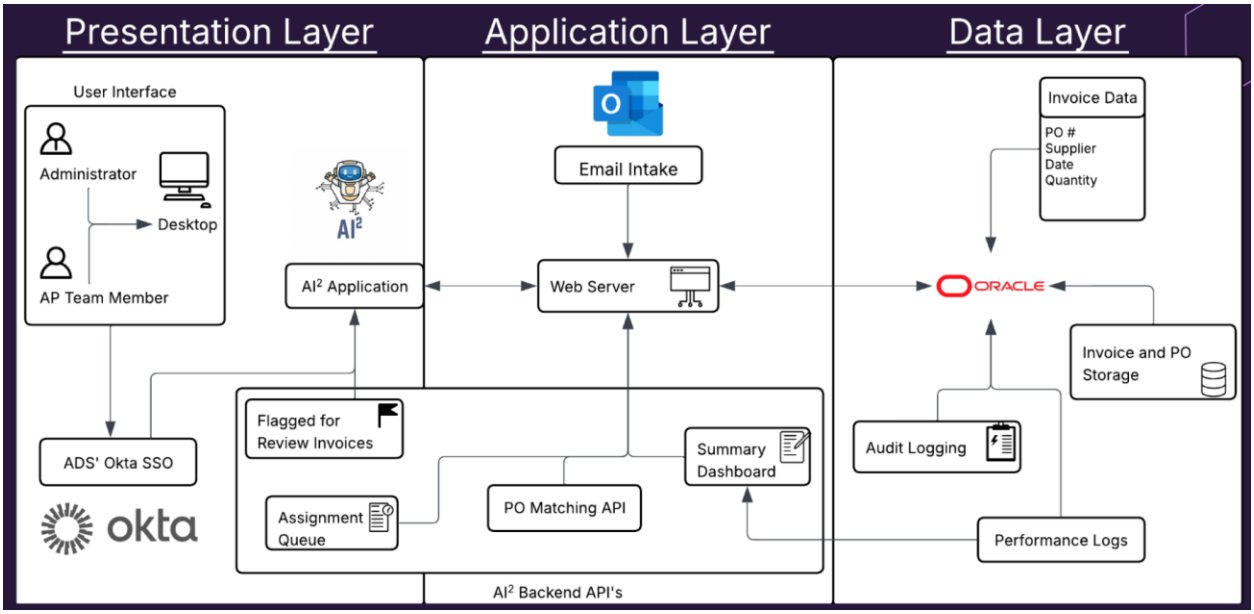
AI^2 is designed for desktop computer use only. All users are authenticated through Okta SSO to ensure secure access and compliance. AI^2 connects to an Outlook inbox using Microsoft Graph API to pull incoming invoices, and to Oracle NetSuite through the Alteryx API to process the necessary data. From the desktop, authenticated users will view the presentation layer. The main components are the assignment queue and the low-confidence invoices that have been flagged for review.

The application layer handles the invoice extraction from the e-mails and transferring the results to the data layer and to the AI^2 backend. The backend is where the matching API will attempt to match an invoice to a purchase order and give a confidence score. AI model trainers will view the summary dashboard, which shows an assessment of the model's performance in extracting the necessary invoice fields and finding a PO match.

The data layer stores all extracted invoice and PO data, as well as an immutable audit log indicating changes to the AI model and the database. All major functional components are shown in Figure 1.

Figure 1

Major Functional Components Diagram



3 Identification of Case Study

The main users for AI² are companies that still perform AP manual processing and are seeking to automate their workload. AI² provides a unique toolset for AP team members, AI model trainers, administrators, and compliance officers to work together and ensure a smooth workflow.

First, administrators will configure AI² as needed to suit the company's workflow. This includes connecting AI² to the company's existing sign-on systems, and to the email inbox which receives invoices. AP team members will experience the largest change to their workflow, with a large volume of high-confidence, low-risk purchase order matches being automated. Employees who were previously tasked with manually matching up to 100 invoices and purchase orders daily will be able to focus on handling high-risk invoices and correcting errors. AI model trainers will monitor AI²'s performance over time and make changes as needed to improve its invoice parsing and matching capabilities. Finally, compliance officers are able to view a log of every

action taken on the system and generate reports to ensure that AI^2 adheres to company policy and meets all regulatory requirements.

The primary goal of AI^2 is to handle the large volume of invoices that a company like ADS receives and processes. To assess its ability to integrate into existing workflows and lighten the AP team's load, a case study will be performed at a company that has a large invoice intake. Administrators will first integrate AI^2 with the company's existing security systems, which will likely be slightly different for each company, and connect it to the email inbox. The incoming invoices will be in a variety of formats, testing AI^2's ability to parse data in unfamiliar forms. It may also make matches with low confidence scores, flagging them for the AP team to review and correct as needed. Performance can be assessed through the number of correct matches and the number of invoices the AP team can process each day. The AI model trainers can assess the performance and make changes as needed, with the hope of seeing improvements over time. Compliance officers will be able to generate all necessary reports to ensure that actions taken through AI^2 align with company policy and external regulation.

At full functionality, AI^2 will become fully integrated into the company's working environment, connecting to a single sign-on system to give each user access to the tools their job requires. It will be able to extract data from a wide variety of formats, help the AP team handle high volumes of invoices, and report any errors to the system administrators. Through further refinement, it will be able to securely integrate into more clients' authentication systems, according to their individual security requirements.

4 Glossary

AI² (Artificially Intelligent Invoices): The proposed automated invoice processing system designed to streamline Accounts Payable workflows through AI-driven matching, classification, and routing.

Alteryx: A data automation and analytics tool that enables secure API connections and data workflows between applications.

AP (Accounts Payable): The amount a business owes for goods and services purchased on credit; typically due at intervals of 30, 45, 60, or 90 days depending on repayment terms.

APEX: A strongly typed, object-oriented programming language that Salesforce developers use to execute flow and transaction control statements on the Salesforce platform.

Atlantic Diving Supply (ADS): An American federal contractor company that provides equipment and logistics solutions to the Department of Defense, federal agencies, and first responders.

AWS GovCloud: A secure U.S. government-compliant cloud environment that supports sensitive data and regulated workloads in accordance with FedRAMP and DoD requirements.

CMMC (Cybersecurity Maturity Model Certification): A framework developed by the U.S. Department of Defense to assess and enhance the cybersecurity posture of contractors handling federal information.

ERP (Enterprise Resource Planning System): An integrated software system used to manage core business processes, such as finance, procurement, and supply chain operations.

FedRAMP (Federal Risk and Authorization Management Program): A U.S. government program that standardizes security assessment, authorization, and continuous monitoring for cloud services.

Graph API: A Microsoft API interface used to access and automate data operations within Outlook and other Microsoft 365 services.

Invoice: A document listing goods or services provided, including a statement of the amount due; a bill.

Oracle: The system of record used by ADS for enterprise data and financial management.

PO (Purchase Order): The official order list ADS sends to suppliers detailing the items or services requested.

Salesforce: A robust Customer Relationship Management (CRM) platform that allows businesses to manage customer relationships efficiently.

Supplier: A person or organization that provides goods or services needed by another organization.

Vendor: A person or company offering goods or services for sale.

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